Huawei Access Node YEL-ACC-52 Optical Transport Failure Alarm ALARM_3006 SOP

Node: YEL-ACC-52 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_3006

Description

The Huawei Access Node YEL-ACC-52 has reported a critical Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) alarm with alarm code ALARM_3006. This alarm is indicative of a critical failure in the optical transport domain, which is a critical component of the Access Node's functionality. The Access Node uses Huawei's Optical Transport Network (OTN) technology to transport client signals over optical fibers. The OTN technology utilizes SFP/XFP transceivers to convert electrical signals into optical signals for transmission over fibers. The Rx/Tx power loss and SFP/XFP degradation are indicative of a failure in the optical transport path, which can be caused by various factors such as fiber cuts, SFP/XFP degradation, or optical signal quality degradation. This failure can impact the reliability and availability of the Access Node and the overall network. Therefore, it is essential to investigate and resolve this issue promptly.

Pre-requisite Checks

- Verify that the Access Node is powered on and all fans are spinning.

- Check the site location and confirm that it is the Yelahanka Sub-station.

- Verify that the redundant link is operational and not affected by the issue.

- Dispatch remote hands to the Yelahanka Sub-station to assist with the troubleshooting and repair process.

Potential Root Causes

Cause	Probability
Fiber Cut	High
SFP/XFP Degradation	Medium
Rx/Tx Power Loss	Medium
Optical Signal Quality Degradation	Low

Diagnostic Commands & Expected Outputs

```
Command: display optical-transceiver, Expected Output: Transceiver Information: SFP/XFP type, Serial number, Voltage, Temperature, and Optical Power., Description: This command displays the transceiver information, which can help identify if the SFP/XFP is degraded or faulty.
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```
Command: display optical-link, Expected Output: Link Information: Fiber type, Distance, and Optical Power., Description: This command displays the link information, which can help identify if the fiber is cut or the optical power is degraded.

Command: display rx-tx-power, Expected Output: Rx/Tx Power Information: Received and Transmitted power levels., Description: This command displays the Rx/Tx power information, which can help identify if the power levels are within the normal range or degraded.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP transceiver, Command: replace optical-transceiver, Description: Replace the SFP/XFP transceiver with a new one to resolve the issue.

Step 2: Step: Re-align the fiber, Command: realign optical-fiber, Description: Re-align the fiber to ensure that it is properly connected to the SFP/XFP transceiver.

Step 3: Step: Adjust the Rx/Tx power levels, Command: adjust rx-tx-power, Description: Adjust the Rx/Tx power levels to ensure that they are within the normal range.

Post-Check Verification

```
Command: display optical-transceiver, Expected Output: Transceiver Information: SFP/XFP type, Serial number, Voltage, Temperature, and Optical Power., Description: Verify that the transceiver information is normal and the SFP/XFP is functioning properly.

Command: display optical-link, Expected Output: Link Information: Fiber type, Distance, and Optical Power., Description: Verify that the link information is normal and the fiber is properly connected.

Command: display rx-tx-power, Expected Output: Rx/Tx Power Information: Received and Transmitted power levels., Description: Verify that the Rx/Tx power information is normal and the power levels are within the normal range.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP transceiver, Command: restore optical-transceiver, Description: Restore the original SFP/XFP transceiver to its previous state.

- Step: Revert the fiber alignment, Command: revert optical-fiber, Description: Revert the fiber alignment to its previous state.

- Step: Revert the Rx/Tx power levels, Command: revert rx-tx-power, Description: Revert the Rx/Tx power levels to their previous state.